Chapter-2

Microprocessor Architecture and Microcomputer System

Microprocessor

- A microprocessor is programmable logic device, designed with registers, flip-flops, and timing elements. It is a single chip made by LSI or VLSI technology. It is the CPU of a microcomputer.

Microprocessor Unit (MPU)

- In early processors the different units of a CPU are made not in a single chip rather in different separate chips.These group of chips are called a Micro Processor Unit (MPU).

2.1 MICROPROCESSOR ARCHITECTURE AND ITS OPERATIONS

Architecture of Microprocessor

- The internal logic diagram of a microprocessor is called its architecture.

Different Categories of Functions Performed by a Microprocessor

All the various functions performed by the microprocessor can be classified in three categories:

- Microprocessor-initiated operations.
- Internal data operations.
- Peripheral (or externally) initiated operations.

2.1.1 Microprocessor-Initiated Operations

The MPU performs primarily four operations:

- Memory Read: Reads data from memory.
- Memory Write: Writes data into memory.
- I/O Read: Accepts data from input devices.
- I/O Write: Sends data to output devices.

Steps to Communicate with Peripherals or Memory Location

To communicate with a peripheral or a memory location, the MPU needs to perform the following steps:

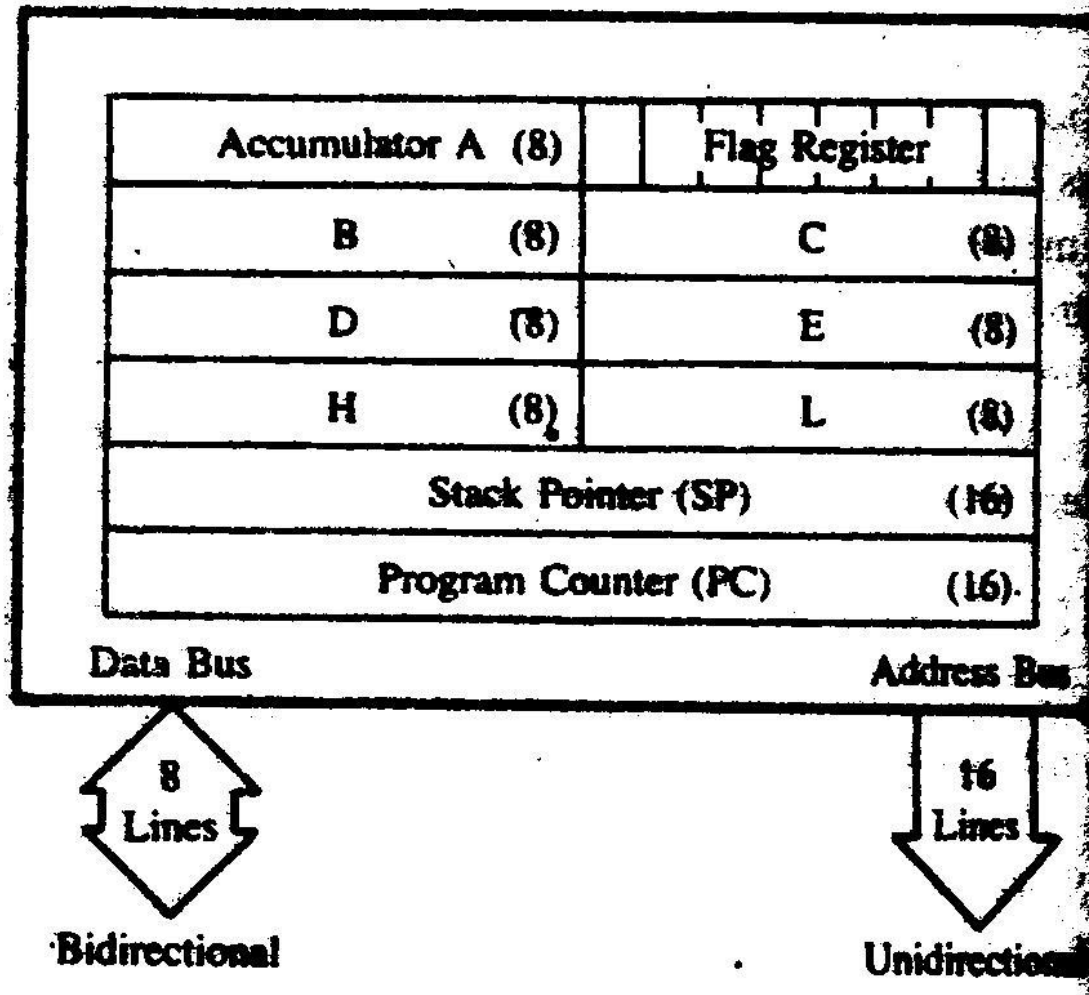
- Step 1: Identify the peripheral or the memory location (with its address)
- Step 2: Transfer data.
- Step 3: Provide timing or synchronization signals.

2.1.2 Internal Data Operations and the 8085/8080A Registers

The internal architecture of the 8085/8080A microprocessor determines how and what operations can be performed with data. These operations are:

- Store 8-bit data
- Perform arithmetic and logical operations.
- Test for conditions.
- Sequence the execution of instructions.
- Store data temporarily during execution in the defined R/W memory locations called the stack.

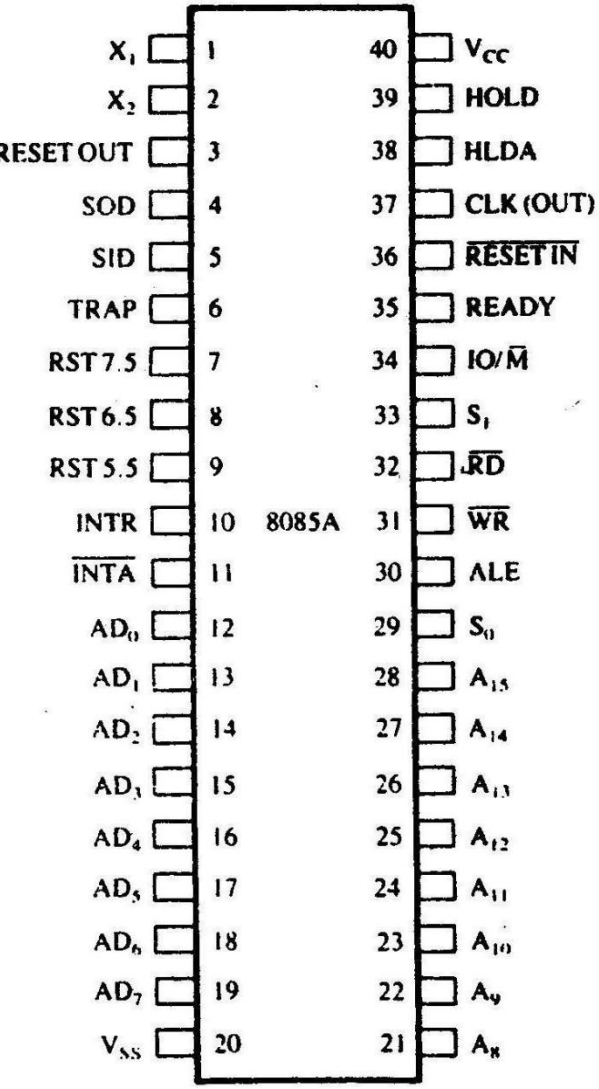
FIGURE 2.3
The 8085 Programmable Registers



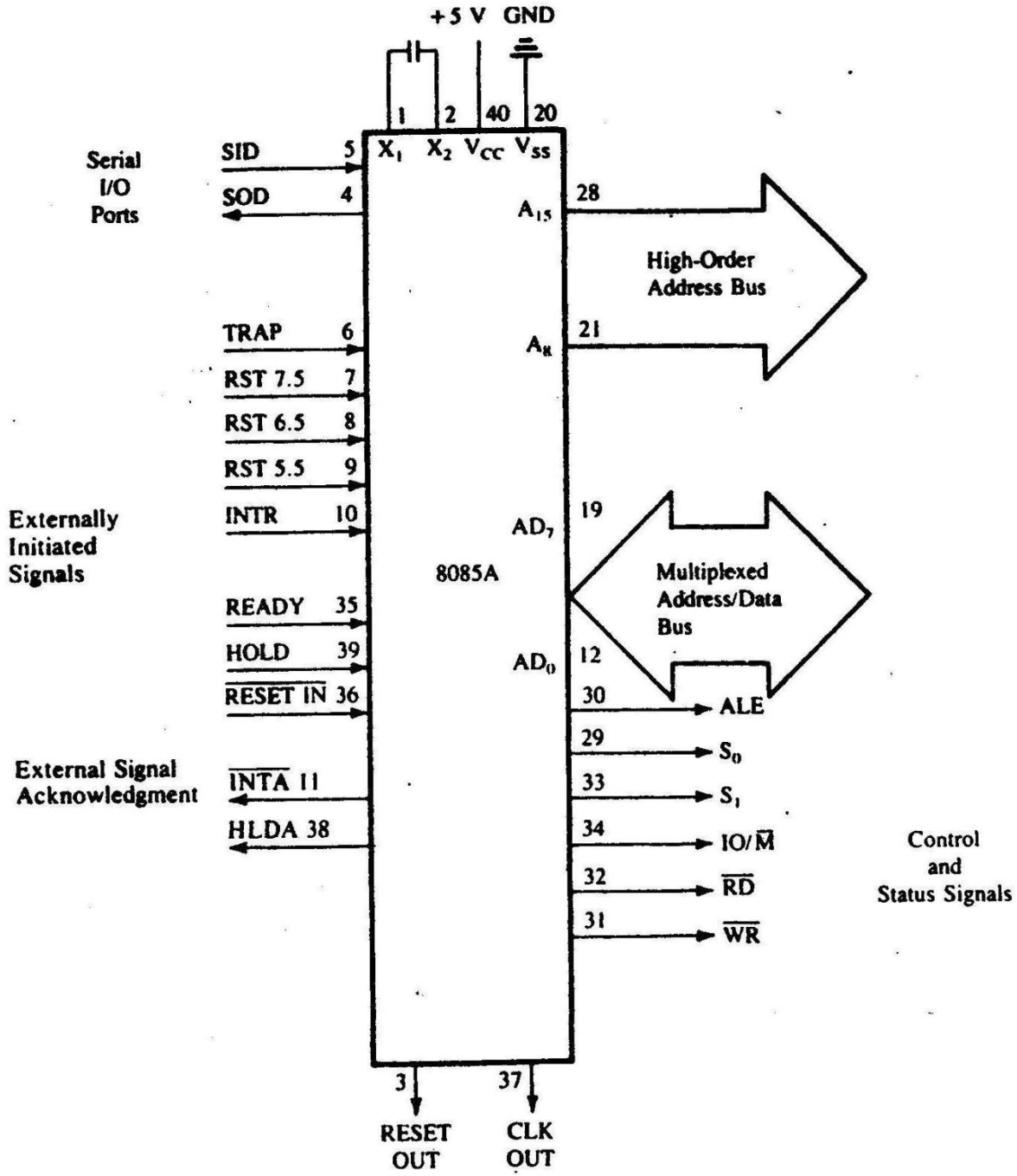
2.1.3 Peripheral or Externally Initiated Operations

External devices (or signals) can initiate the following operations for which individual pins on the microprocessor chip are assigned:

- **Reset:** When all internal operations are suspended and the program counter(PC) is cleared (0000H).
- **Interrupt:** Microprocessor can be interrupted from the normal execution and asked to execute some other instructions called service routine. After completing service routine it resumes its operation.
- **Ready:** If the signal of 8085/8080A READY pin is low, the microprocessor enters into a Wait state. The pin is used to synchronize with slower peripherals with microprocessor.
- **Hold:** When the HOLD is activated by an external signal, the microprocessor give up control of system bus for external peripheral to use them. For example, HOLD signal is used in DMA data transfer.


X1 1
X2 2
RESET OUT 3
SOD 4
SID 5
TRAP 6
RST 7.5 7
RST 6.5 8
RST 5.5 9
INTR 10
INTA 11
AD0 12
AD1 13
AD2 14
AD3 15
AD4 16
AD5 17
AD6 18
AD7 19
VSS 20
8085A
40 VCC
39 HOLD
38 HLDA
37 CLK (OUT)
36 RESET IN
35 READY
34 IO/M
33 S1
32 RD
31 WR
30 ALE
29 S0
28 A15
27 A14
26 A13
25 A12
24 A11
23 A10
22 A9
21 A8

8085 Pinout


+5 V
GND
1
2
40
20
X1
X2
VCC
VSS
Serial I/O Ports
SID 5
SOD 4
A15 28
High-Order Address Bus
A8 21
TRAP 6
RST 7.5 7
RST 6.5 8
RST 5.5 9
INTR 10
Externally Initiated Signals
AD7 19
8085A
Multiplexed Address/Data Bus
READY 35
HOLD 39
RESET IN 36
AD0 12
30 ALE
29 S0
33 S1
34 IO/M
32 RD
31 WR
Control and Status Signals
External Signal Acknowledgment
INTA 11
HLDA 38
3
37
RESET OUT
CLK OUT